

Philipp Kleer

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ABOUT ME

Data Scientist (PhD) with **8+ years of experience** in **quantitative research**, statistical modeling, and **data-driven decision support**. Expert in bridging the gap between **complex statistical modeling** and **production-grade MLOps**, translating complex analyses into **actionable insights for diverse stakeholders**. Experienced in **international and interdisciplinary environments**.

WORK EXPERIENCE

Data Scientist

Center for Studies and Research in Network Technology and Operations - Measurements (CEPTRO), [NIC.br](#)

July 2025 — Present

São Paulo, Brazil

NIC.br is the executive arm of the Brazilian Internet Steering Committee, managing national-level network data

- **Built and productionized** a classification system for internet connection type detection, integrating **MLflow** for experiment tracking and model versioning
- **Analyzed** and **optimized** large-scale **PySpark** ETL pipelines
- **Architected** and migrated monolithic applications (R/Shiny) to a modular micro-services structure, reducing codebase complexity by 40% and cutting deployment lead time by 65% through automated CI/CD pipelines, Docker and caching
- **Designed** and maintained **data pipelines, databases**, and APIs (PostgreSQL/PostGIS), making **architectural decisions** on where to execute operations (database vs. analytics layer) and optimizing performance and data transfer (Parquet, efficient APIs) in production analytics portals leading to faster data availibilites in the applications
- **Implemented CI/CD pipelines** (GitLab) and reproducible environments (renv, uv), integrating automated testing, linting, and deployment of containerized applications with **Docker** and **Kubernetes** shrinking deployment time significantly
- **Developed interactive dashboards and data visualizations** (ggplot2, plotnine, ggiraph, plotly) with standardized design and narrative reporting for technical and political audiences.

Research Scientist & Lecturer

Justus-Liebig-University Giessen

April 2025 — July 2025

Giessen, Germany

- Authored **3 publications/chapters**, including **applied material on teaching R** to social science students.
- **Designed and taught 4 courses** (~ 200 students)

Research Scientist

Justus-Liebig-University Giessen

February 2023 — September 2024

Giessen, Germany

- Directed **end-to-end data research initiatives**, translating **complex statistical models into actionable strategic reports** for international policy-makers
- Used SEM and ML methods to provide insights for international policy-makers (**post-double selection**, EFA/CFA, SEM, multilevel models, regression)
- Produced **8 publications** and technical reports, **translating complex analyses into actionable insights**.
- Collaborated in **international research teams** and conducted research stays in Brazil and Spain.

Doctoral fellow

Justus-Liebig-University Giessen

October 2015 — September 2022

Giessen, Germany

- Conducted **quantitative research** (survey based) using **R**, **Git**, and **reproducible workflows**.
- Applied **advanced statistical modeling techniques**, including multilevel models, SEM, Bayesian regression, and quantitative text analysis.
- Published **6 peer-reviewed papers/reports** and presented findings at **9 international conferences**.
- Received **advanced training in machine learning with R**, **Bayesian modeling**, **multilevel regression**, and **quantitative text analysis**.

Project assistant (workshop management)

Justus-Liebig University Giessen

May 2019 — May 2022

Giessen, Germany

- Designed and maintained a **SQL-based database** to **automate reporting** and **certification workflows**, significantly reducing manual effort.
- Coordinated large academic events and supervised student assistants, ensuring on-time delivery and quality standards.

SKILLS

Data Science & Analytics: Statistical Modeling | Bayesian Regression | Machine Learning | Feature Engineering | Model validation and selection | Explorative Analysis | Quantitative Research |

Programming & Data: Python (polars, PySpark, scikit-learn, PyTorch, TensorFlow) | R (tidyverse, tidymodels, Shiny, ggplot2) | SQL (PostgreSQL) |

Data Visualization & Reporting: Shiny Dashboards | plotnine | ggplot2 | ggiraph | plotly | Quarto | Interactive Reporting |

MLOps & Engineering: MLflow | Airflow | CI/CD (GitLab) | Docker | Kubernetes | renv | uv |

Version Control & Collaboration: Git | GitHub | GitLab | Code Review | Agile Development |

Languages: English (fluent, C1) | Portuguese (fluent, B2/C1) | German (native, C2)

PROJECTS

- **Fiber Detection MLOps Pipeline:** I developed and deployed a production-ready binary classification system by designing a precision-focused modeling workflow (including threshold tuning and probability calibration) and implementing a Kubernetes-native MLOps pipeline (CronJobs + self-hosted MLflow), enabling automated retraining, controlled model promotion (Champion/Challenger), and reproducible deployments under constrained infrastructure conditions.
- **Connectivity in Health:** I redesigned a Shiny application in production by optimizing data pipelines (Parquet + polars), validating performance under load, and implementing reproducible builds with multi-stage Docker, renv, and CI/CD, ensuring operational predictability and sustainable maintenance.
- **Shiny application Observatory of Artificial Intelligence (OBIA):** Refactored a large production Shiny application into a modular, testable architecture, reducing code size by ~41%. Implemented CI/CD, automated testing, linting, and reproducible environments (renv) to support reliable multi-developer collaboration.
- **Shiny application Self-Diagnosis of the Information Technology Resource Management System:** Developed and deployed a production-grade Shiny dashboard assessing IT governance among national entities, including data integration, standardized visual design, and CI/CD-based deployment for public-facing analytics.
- **Network Technology Analysis & Presentation:** Conducted statistical and exploratory analyses of network technologies, translating results into high-impact presentation for technical and policy stakeholders at IX Forum 2025.
- **ML Ops Pipeline Project:** Implemented an end-to-end MLOps workflow for model training, versioning, and monitoring using MLflow and DVC, simulating production-like model lifecycle management.

EDUCATION

Doctorate (Dr. rer. soc.) Justus-Liebig University Giessen	May 2016 — December 2022 Giessen, Germany
• degree in Social Sciences with specialization on quantitative methods; doctorate was accomplished at the Chair of Empirical Research Methods	
Master of Arts University of Vienna	October 2011 — May 2014 Vienna, Austria
• degree in Political Science with a focus on quantitative methods and Central Eastern European societies	
Bachelor of Science University of Cologne	October 2008 — August 2011 Cologne, Germany
• degree in Social Sciences with a strong emphasis on empirical research methods	

CERTIFICATES

Data Science & MLOps DataScientest (certified by Panthéon-Sorbonne, in total 505 hours)	October 2024 — March 2025 Remote
• Completed a 505-hour certified Data Science & MLOps program (Panthéon-Sorbonne).	
• Built machine learning models for classification, regression, and clustering using <code>scikit-learn</code> , TensorFlow, and PyTorch.	
• Developed and tracked ML pipelines with MLflow , DVC , Dagshub , Airflow , and ZenML .	
• Deployed models using BentoML and gained hands-on experience with Docker , Kubernetes , Prometheus , and Grafana .	
• Created analytical visualizations and interactive apps with matplotlib, seaborn, plotly, and Streamlit.	